

# Ioanna Vavelidou

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|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EDUCATION           | <b>PhD in Electrical Engineering</b><br>Stanford University, CA<br>Advisor: Caroline Trippel, High Assurance Computer Architectures Lab<br><i>Sept 2021 - Present</i>                                                                                                                                                                                                                                                                                                                              |
|                     | <b>MS in Electrical Engineering</b><br>Stanford University, CA<br><i>Sept 2022 - June 2024</i>                                                                                                                                                                                                                                                                                                                                                                                                     |
|                     | <b>Diploma in Electrical and Computer Engineering (Joint BSc &amp; MEng)</b><br>National Technical University of Athens, Greece<br>Advisor: Professor Nectarios Koziris, Computing Systems Laboratory<br>Thesis: <i>Efficient Shared Resource Management in Multiprocessors using Reinforcement Learning techniques</i><br>GPA: 9.26/10 (top 3%)<br>Department Entrance Rank: 1st (Highest score in the National University Admission Exams, top 0.01% nation-wide)<br><i>Oct 2015 - July 2021</i> |
| RESEARCH EXPERIENCE | <b>Ph.D Research Assistant</b><br>Stanford University, CA<br>Advised by Professor Caroline Trippel<br>Topic: <i>Sustainability-Aware Hardware Reliability in the Datacenter</i><br><i>April 2022 - Present</i>                                                                                                                                                                                                                                                                                     |
|                     | <b>Undergraduate Researcher</b><br>National Technical University of Athens, Greece<br>Thesis: <i>Efficient Shared Resource Management in Multiprocessors using Reinforcement Learning techniques</i><br><i>Dec 2019 - July 2021</i>                                                                                                                                                                                                                                                                |
| INTERNSHIPS         | <b>Student Researcher at Google Platforms</b><br><i>Developing an ML-driven probability model for predicting insidious hardware failures in Google's AI chip fleet, enabling smarter testing and performance trade-off decisions.</i><br><i>Feb 2026 - June 2026</i>                                                                                                                                                                                                                               |
|                     | <b>Student Researcher at Google DeepMind</b><br><i>Deployed and evaluated real-world ML workloads as hardware reliability tests, identifying which workloads best expose defect-induced errors in AI accelerators.</i><br><i>June 2025 - Dec 2025</i>                                                                                                                                                                                                                                              |
|                     | <b>Student Researcher at Google Platforms</b><br><i>Designed compiler-based functional tests for Silent Data Corruption detection in CPUs.</i><br><i>October 2023 - June 2024</i>                                                                                                                                                                                                                                                                                                                  |
|                     | <b>PhD Software Engineer Intern at Meta Platforms</b><br><i>Optimized memory usage for ML workloads to improve performance and scalability</i><br><i>June 2022 - Sept 2022</i>                                                                                                                                                                                                                                                                                                                     |
| HONORS & AWARDS     | <b>Award by the President of the Republic of Greece</b><br><i>For achieving the highest National University Admission score in my educational district (Attica district A')</i><br><b>Award and Grant by the National Technical University of Athens</b><br><i>For achieving the highest entrance score in my department</i><br><b>Award and Grant by Eurobank EFG, "The Great Moment for Education"</b><br><i>For achieving the highest National University Admission score in my High School</i> |
| PUBLICATIONS        | <b>Ioanna Vavelidou, Subho Banerjee, Eric X. Liu, Mike Fuller, Subhasish Mitra, Caroline Trippel</b><br><b>ITHICA: Intra-Thread Instruction Checking Approach for Defect-Induced</b>                                                                                                                                                                                                                                                                                                               |

**Data Corruptions**  
(in submission)

Saranyu Chattopadhyay, Keerthikumara Devarajegowda, Bihan Zhao, Florian Lonsing, Brandon Alan D’Agostino, **Ioanna Vavelidou**, Vijay Deep Bhatt, Sebastian Prebeck, Wolfgang Ecker, Caroline Trippel, Clark Barrett, Subhasish Mitra

**G-QED: Generalized QED Pre-silicon Verification beyond Non-Interfering Hardware Accelerators**

*DAC 2023, Design Automation Conference*

Hang Qiu, **Ioanna Vavelidou**, Jian Li, Evgenya Pergament, Pete Warden, Sandeep Chinchali, Zain Asgar, Sachin Katti

**ML-EXRAY: Visibility into ML Deployment on the Edge**

*MLSys 2022, Conference on Machine Learning and Systems*

SKILLS

**Languages:** English (highly proficient, Michigan ECPE C2 Degree, TOEFL), French (adequate working knowledge, Delf B2 Degree), Greek (native)

**Programming Languages:** Python, C, C++, MATLAB, x86 Assembly, LLVM IR

**Relevant Coursework:** Machine Learning on Embedded Systems, Reinforcement Learning, Human-Computer Interaction Research, Cloud Computing Research, Principles of Data-Intensive Systems, Performance Engineering of Computer Systems & Networks, Computer Systems Architecture (**Stanford**), Speech and Natural Language Processing, Pattern Recognition, Computer Vision, Advanced Robotics, Advanced Topics in Computer Architecture, Logic Design of Digital Systems, Operating Systems, Networks and Circuitry Theory, VLSI System Design (**NTUA**)

GRE SCORES

GRE General Aug. 2020: Quantitative Reasoning 169/170, Verbal Reasoning 163/170, Analytical Writing 4.5/6

PROFESSIONAL  
MEMBERSHIPS

- ACM Member, ACM SIGARCH
- Computing Research Association - Widening Participation (CRA-WP)
- IEEE Member